



Examiners' Report Summer 2026

Pearson Edexcel GCE

In Further Mathematics (9FM0)

Paper 4A Further Pure Mathematics 2

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General comments

This paper included a mixture of questions on topics covered in the specification and questions requiring problem solving. The first four questions included accessible parts, while the later questions involved greater challenge. Question 7, on reducible integration, may have been perceived as challenging; however, it was broken down into stages, which supported learners in demonstrating their understanding and using earlier parts to continue their solutions.

The paper provided broad coverage of the specification, and responses suggest that learners were generally able to complete it within the time available.

For combinatorics questions, learners are reminded to show the possible sets, values and combinations clearly. Systematic listing helps examiners understand the reasoning and can also help learners identify whether any cases have been missed.

Comments on individual questions

Question 1

Learners generally found this to be a familiar opening question on finding the length of a curve, using the formula provided in the formula booklet.

Responses to this question were often successful, with many learners gaining full marks. Some responses used an incorrect formula or integrated the given curve directly. Where learners attempted integration by substitution, marks were sometimes not awarded due to errors when replacing the original variable with the substitution variable, or through not adjusting the boundary values correctly.

Question 2

Parts a and b were answered with a high degree of success, aside from small numeric slips.

Part (a)

A common error was $\frac{10!}{4!}$ instead of ${}^{10}P_4$

Part (b)

Different powers of 10 were seen

Part (c)

In part (c)(i), some responses were difficult to follow because the method used to count the different possibilities was not always clear. More successful responses tended to list the outcomes systematically, making it easier to see how each possibility had been counted.

The need for a systematic approach was also evident in part (c)(ii). Stronger responses clearly listed the combinations that summed to 34, 35 and 36, then subtracted these combinations, together with the answer to part (c)(i), from their answer to part (b). Some responses missed possible outcomes, and it was not always clear which outcome was associated with each count.

Learners are advised to list clearly the numbers used for each sum and to show how the number of possible combinations has been found. This helps examiners follow the working and can also help learners identify whether any cases have been missed.

Question 3

Part (a)

Stronger responses made it clear the connection between 75% and $\frac{3}{4}h_n$ and 20cm and $\frac{1}{5}$. Some responses did not mention 75% or 20cm. Most learners gained both marks clearly explaining where the model has come from.

Part (b)

Most learners solved the particular solution first, then used that as part of the general solution using CF + PS with the initial $h_0 = 1$ condition to find the value of a constant and tended to gain full marks. The incorrect particular solution $A\lambda + b$ or turning it into a quadratic equation were seen.

Learners who attempted to use a geometric series approach often started with an incorrect form $h_n = h_0 \left(\frac{3}{4}\right) + \sum_{r=1}^n \left(\frac{1}{5}\right)$ scoring no marks.

Part (c)

Those learners who found a correct answer to part (b) could generally interpret their solution correctly. Units were required for this part, there were many answers of 0.8 but with no units or 0.8 cm, which scored B0.

Part (c)

Learners need to give a limitation of the model and look where the modelling appears in the question. Most responses used mechanics limitations, e.g. air resistance, friction, dimension of the ball etc. which scored B0. The modelling is with the step, and 75% height is reached each time, so a good assumption is there will not be infinitely many steps, or it will not reach 75% of the previous height each time.

Question 4

This question tested knowledge and understanding of number theory. Part (i) was standard processing which learners understood well. Parts (ii) and (iii) involved a bit more problem solving and applying theorems.

Part (i)(a)

Most responses proceeded with applying the Euclidean algorithm to find the highest common factor of two numbers, although a notable number made numeric slips leading to highest common factor of 2.

Part (i)(b)

Learners were able to use their solution to (a) find the required integer values. If full marks scored in (a) then also in (b).

Part (ii)

This question was answered well. In stronger responses, learners multiplied both sides of the congruence equation by 2, however, in some responses both sides were multiplied by a different number which involved more simplification to reach the solution. Some learners did not gain the final mark by leaving the answer as $8 \pmod{5}$, or omitting the modulus and just stating that $x = 3$

Part (iii)

Learners proceeded by trying to see if $40^{97} + 97^{40}$ was divisible by 41 and their solutions were varied in terms of answer length and learner success. Most learners showed evidence that they knew and could apply Fermat's Little Theorem to $97^{40} \equiv 1 \pmod{41}$, but there were a wide variety of approaches to evaluating 40^{97} . The stronger responses noticed that $40^{97} = 40^{40 \times 2 + 17} \equiv 40^{17} \pmod{41}$ leading to $(-1)^{17} \pmod{41}$. Some responses stopped at $40^{17} \pmod{41}$ and did not know what to do next. An alternative approach seen was $40^{40} \equiv 1 \pmod{41}$, $40^2 \equiv 1 \pmod{41} \therefore 40^{97} = 40^{40} \times (40^2)^{28} \times 40 \equiv 40 \pmod{41}$. Learners who managed to find both terms as mod 41 then combined into one modulus expression, either $(-1)^{17} + 1 \equiv 0 \pmod{41}$ or $40 + 1 \equiv 41 \pmod{41}$. Having reached this far an appropriate conclusion was stated.

Question 5

Part (a)

Part (a) was generally answered very well, and the most common approach was factorising the characteristic equation and solving; this led to the correct values for a and b . Some learners used their calculator to find the roots but then did not always make the correct connection between these roots, 1 and -2, and the corresponding values of a and b .

Part (b)

Part (b) involved quite a lot of work using the Cayley-Hamilton theorem to get the first two marks, but this was successfully completed by most learners. Many

solutions did not score any more marks than this, though, despite getting to the

correct answer, because they attempted to solve $\mathbf{M} \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$ rather than

$\mathbf{M} \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} = \lambda \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix}$. If the learners just assumed that the eigenvalue would be 1,

rather than using a variable or fully testing both possibilities, then they scored no further marks for this part. Using eigenvalues of 1 and 2, finding the value of k in each case and selecting the correct value could gain the marks.

Question 6

This question tested learners' understanding of transformations involving complex numbers. Some responses suggested that this was a less familiar presentation of the topic, and learners did not always identify a clear starting point.

Part (a)

Part (a) was completed successfully by many learners, although some responses did not present the answer in the required form. They started with $w = e^{2+3i}$ but instead of writing as $w = e^2 e^{3i}$ they found the modulus and argument of $2 + 3i$ first and then had $w = e^{\sqrt{13}e^{0.9826i}}$

Part (b)

In part (b), some responses gained the initial marks but did not fully justify that the image was a unit circle. In these responses, the modulus was not always shown to be 1. Learners are reminded that, in 'show that' questions, conclusions should be fully justified. In stronger responses, reaching $w = e^{iy}$ was followed by stating that since $r = 1$, or equivalent, it is a unit circle.

Part (c)

Many learners correctly started with $w = e^x e^{\frac{\pi}{4}i}$ or $w = e^x \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$ following

the structure to part (a). This enabled learners to be able to draw the correct image but did not use the correct vocabulary to describe the image geometrically, a ray (half line) starting at the origin. Some incorrect answers involve rotations of

$\frac{\pi}{4}$ or straight lines.

Question 7

Although this question may have appeared demanding at first, many responses showed success across substantial parts of it. The question was structured in several distinct stages, which supported learners in using earlier results to progress through the later parts.

Part (a)(i)

This was answered well, most responses recognising that they need to evaluate $\int_0^2 \cosh x \, dx$. Occasionally the answer was given as a decimal instead of an exact value.

Part (a)(ii)

Most learners kept the bracketed term together, rather than multiplying out into many separate terms and were generally more successful. Many learners multiplied out the brackets and used integration by parts on two integrals. This could lead, potentially, to sign errors and numerical slips.

Part (b)

As a mechanical exercise in the product and chain rules, most learners gained the first mark in this part. The next mark challenged learners as they needed to spot $(x-1)^2$ factor and multiply out reaching $(\{x^2 - 2x\} + 1)$ and apply to their derivative. Having this result aided learners to see how to approach the question. Learners then managed to score the final marks. Generally, learners scored well on this part.

Part (c)

This was the part of the question most frequently left unattempted, although some responses included fully successful solutions. A successful approach was to use integration by parts to obtain the required intermediate expression

$$I_n = \left[(x^2 - 2x)^n \sinh x \right]_0^2 - \int_0^2 n(x^2 - 2x)^{n-1} (2x - 2) \sinh x \, dx.$$

Applying integration by parts again on the integral was required with $u = (x^2 - 2x)^{n-1} (2x - 2)$ which was the challenging part. This resulted in

$$\left[(x-1)(x^2 - 2x)^{n-1} \cosh x \right]_0^2 - \int_0^2 \left((2n-1)(x^2 - 2x)^{n-1} + 2(n-1)(x^2 - 2x)^{n-2} \right) \cosh x \, dx$$

where the answer to part (b) can then be used.

Some successful responses worked backwards from part (b), multiplying both sides by the relevant factor and then integrating.

Part (d)

This part was generally answered well. Most responses recognised that they needed to apply the result from part (c), together with their answers to part (a).

Question 8

This question was testing the axioms of a group with an operation involving set notation.

Part (a)

Part (a) was generally answered well, although some responses omitted a value from one of the sets.

Part (b)

The identity element was sometimes incorrectly written as $\{0\}$ implying the number 0, when it needed to be the empty set $\emptyset$ or $\{ \}$.

The next mark was to show that the empty set was the identity element as the demand for the question was to show that it is a group. Learners needed to state that $\emptyset \oplus A = A = A \oplus \emptyset$ or $\{ \} \oplus A = A = A \oplus \{ \}$.

Learners should be reminded that, in this type of question, it is important to show explicitly that the empty set is the identity element, for example by giving an equivalent statement involving the operation. Many responses were able to show that each element was self-inverse.

The final mark required a statement about closure, together with reference to the identity and inverse elements, before concluding that the set formed a group. Some responses listed the group axioms but did not apply them to the operation given in the question.

As this was the final question on the paper, the fact that it was attempted by many learners suggests that time pressure was not a significant issue overall.